

Homework[5]:FFT

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1. LINKS TO CODE

Here is a link to the code file [HW5 fft.ipynb](#).

2. QUESTION 1:

2.1. Part a:

$$\begin{aligned}\hat{f}(k) &= \int_{-\infty}^{\infty} e^{-50(x-\frac{1}{2})^2} e^{-ikx} dx \\ &= \int_{-\infty}^{\infty} e^{-50(x^2-x+\frac{1}{4})} e^{-ikx} dx \\ &= \int_{-\infty}^{\infty} e^{-50x^2+50x-\frac{50}{4}} e^{-ikx} dx \\ &= e^{-25/2} \int_{-\infty}^{\infty} e^{-50x^2+(50-ik)x} dx.\end{aligned}$$

Then using equation (2),

$$\int_{-\infty}^{\infty} e^{-ax^2+bx} dx = e^{\frac{b^2}{4a}} \sqrt{\frac{\pi}{a}}$$

we have:

$$\begin{aligned}\hat{f}(k) &= e^{-25/2} \exp\left(\frac{(50-ik)^2}{4 \cdot 50}\right) \sqrt{\frac{\pi}{50}} \\ &= e^{-25/2} \exp\left(\frac{2500-100ik-k^2}{200}\right) \sqrt{\frac{\pi}{50}} \\ &= \exp\left(-\frac{25}{2} + \frac{2500-100ik-k^2}{200}\right) \sqrt{\frac{\pi}{50}} \\ &= \exp\left(-\frac{k^2}{200} - \frac{ik}{2}\right) \sqrt{\frac{\pi}{50}} \\ &= \sqrt{\frac{\pi}{50}} e^{-k^2/200} e^{-ik/2}.\end{aligned}$$

This show that a Gaussian expression remains a Gaussian under the Fourier transform, with only a difference in phase factor.

2.2. Part b:

At three different values of N, on the interval [0,1], we sample the Gaussian and implemented the DFT directly

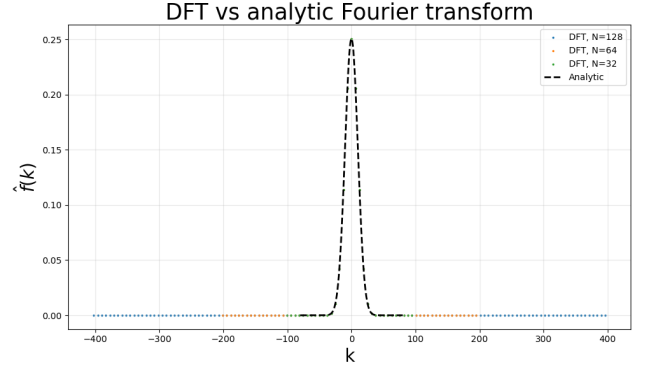


Figure 1. DFT vs Analytic result at 3 different N values

and compared it to the analytic transformation. The result is shown in Figure 1. The results are similar, but higher order of N provides better approximations and further out on the wings.

We also applied the inverse DFT to the transformed data and compared the result in Figure 2. The reconstruction matches the original function, with a small numerical error, which shows the implementation as consistent.

2.3. Part c:

Here, we use the off-the-shelf FFT method from scipy, and compare to our hand-written DFT. The results overlap, as shown in Figure 3. The key difference here is runtime difference.

After over 100 runs for both methods, the off-the-shelf FFT timing shows variability at low N due to noise, but the trend is still clear: the hand-written DFT times increases much more sharply with N, and the off the shelf method is more stable, as shown in Figure 4

Thus, the FFT method is strongly preferred at large scale, even though it produces the same result as the DFT.

3. QUESTION 2

3.1. Part a:

Here, in Figure 6, is the color plot for the Heat equation, with space and time as axis. We used RK 4 method from HW3, with slight modification for the complex

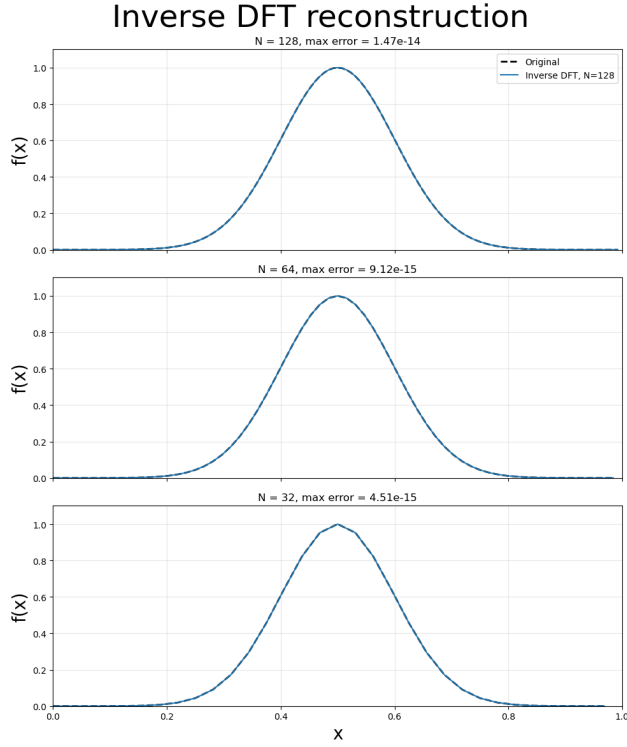


Figure 2. inverse DFT vs Analytic result at 3 different N values

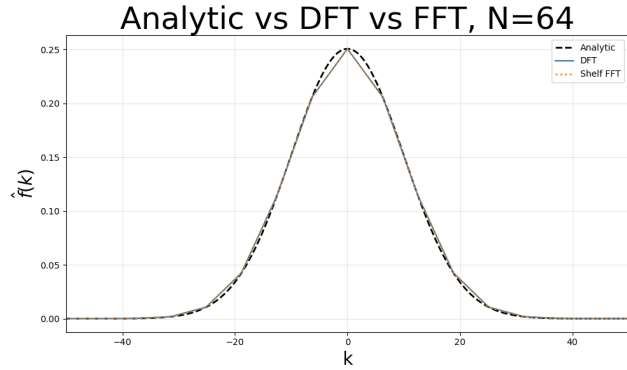


Figure 3. off the shelf FFT vs DFT

plane, to evolve the Fourier mode. We see the initial Gaussian spreads out overtime and decrease in peak intensity.

The total heat remains nearly constant with minor drifts, as shown in Figure ???. The max change was: 4.996×10^{-16} . This makes sense as the average of the solution should be conserved over time.

3.2. Part b:

With a larger a at 0.05 and a larger $\Delta t = 0.01$, we ran into issues with the RK4 method, returning "overflow" or "invalid values" in multiplication. This implied

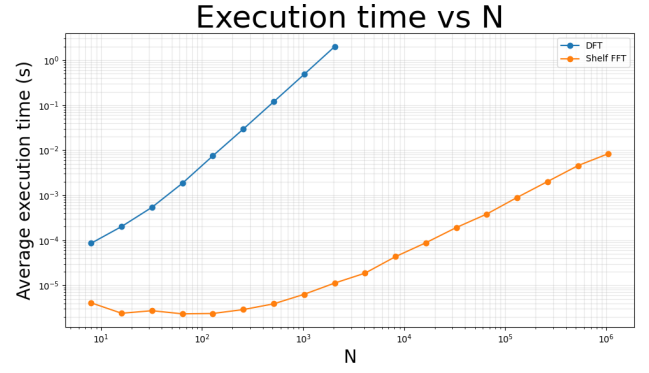


Figure 4. off the shelf FFT vs DFT

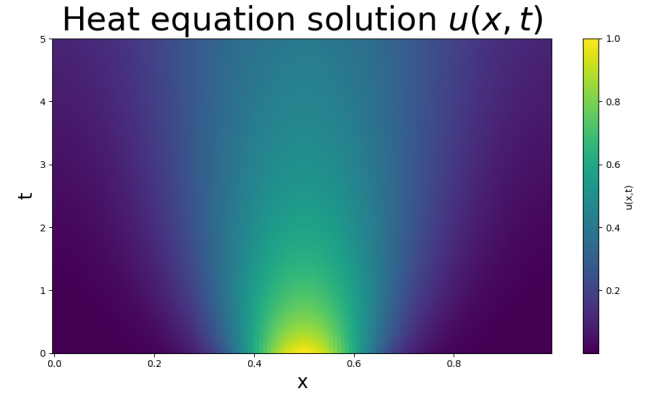


Figure 5. Heat equation color map

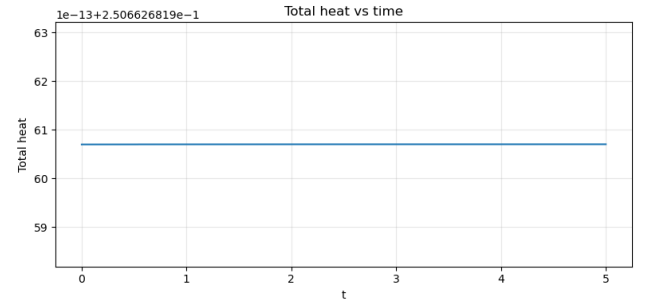


Figure 6. Nearly constant total heat

that the method is unstable at larger timestep and/or diffusivity.

ACKNOWLEDGMENTS

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